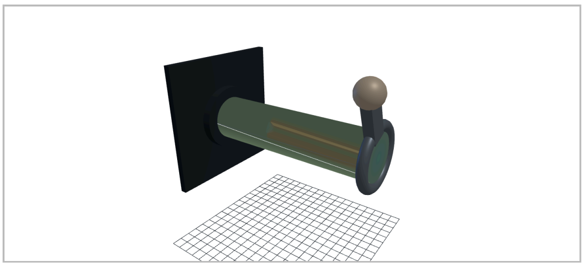
Shaft in Torsion — bench sheet

Ø25 solid × 100 mm · Steel 1045 (cold drawn) · End-milled keyseat (profiled) r 0.50 mm (r/d 0.020)
T = 120.0 N·m at 1450 rpm = 18.22 kW · MechCalc — design check, not a substitute for full analysis

n = 2.61 τ_{max} 117.3 MPa vs 306 MPa allowable · capacity 312.7 N·m · twist 0.23° (limit 0.20°)



The shaft as modelled, shaded by shear stress against $0.577 \cdot \sigma_y$. The hot band sits at the end-milled keyseat (profiled); the end face carries the radial $\tau \propto r$ gradient. Twist shown ×102.

Section & loading

Polar second moment J	**38350 mm⁴**
Section modulus Zp	**3068 mm³**
Shear modulus G	**79.5 GPa**
Nominal surface shear	**39.1 MPa**
Peak shear, Kts 3.00	**117.3 MPa**
Torsional stiffness	**531.8 N·m/°**

Equations used

$\tau = T \cdot c / J = 16T / \pi d^3$ — Surface shear stress (solid shaft)
$J = \pi(d^4 - d_i^4) / 32$ — Polar second moment — hollow or solid
$\tau_{max} = K_{ts} \cdot T \cdot c / J$ — Keyseat, fillet or groove multiplies it
$K_{ts} = K_{ref} \cdot (r/d \div ref)^{-0.238}$ — …and the radius you cut sets Kts
$\tau_{allow} = 0.577 \cdot \sigma_y$ — Distortion-energy shear yield
$\theta = T \cdot L / G \cdot J$, $G = E / 2(1+\nu)$ — Angle of twist and the modulus it works on
$k_t = G \cdot J / L$ — Torsional stiffness, N·m per radian
$T = 9549 \cdot P[kW] / n[rpm]$ — Power ⇄ torque at shaft speed
$n = \tau_{allow} / \tau_{max}$ — Safety factor against shear yield

Static torque only — no bending, no combined stress, no fatigue. Kts is a first-iteration estimate interpolated from Shigley Table 7-1 anchors, not a Peterson chart lookup. Material values are typical reference figures; verify before production use.